PROJECT-BASED LEARNING

PHASE-I EVALUATION

Project Title

Smart Hostel Management & Complaint Resolution System

Team ID: DSCPP-III-2026-T269

Department of Computer Science & Engineering
Graphic Era (Deemed to be University), Dehradun
Academic Session 2026–27

Team & Mentor Details

Team Information

Team ID: DSCPP-III-2026-T269
Semester: 3rd
Section: G, Cyber Security, DS 2
Domain: Web Systems & Automation – *Data Structures & C++*

Team Members

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Mentor

Mrs. Ashwini Yeole

Interactions Completed - 2

Problem Statement & Motivation

Problem Statement

In our college hostels, students note down room complaints in a paper register kept at the warden office. When pages get full or misplaced, complaints get lost. More importantly, urgent safety issues like water leaks or sparking switchboards wait behind basic cleaning chores because complaints are handled strictly by entry order without urgency sorting.

Motivation

- Paper registers get misplaced, damaged, or skipped during staff shift changes.
- Emergency issues (short circuits, pipe burst) get delayed without a priority queue.
- Students have to visit the office repeatedly just to check repair status.

Target Users / Stakeholders

- Primary: Hostel students, floor wardens, and campus repair staff.
- Secondary: Chief warden and hostel administration.
- Expected benefit: Direct digital complaint logging, clear status tracking, and fast emergency repairs.

Objectives

Project Objectives

1. Write C++ OOP classes (Student, Warden, Staff, Room) to model hostel records and room allocations cleanly.
2. Implement a binary heap priority queue so that urgent issues like electrical sparking get handled before routine tasks.
3. Use custom C data structures like Hash Tables for O(1) complaint lookups and FIFO Queues for routine cleaning requests.
4. Store and reload all records using CSV flat files on disk so complaint data stays saved even after the program closes.

Key Objective: Replace slow paper registers with an automated C++ system
that prioritizes hostel complaints by severity and updates repair tasks in real time.

Proposed Solution

Proposed Approach

1. Student & Staff Login Authentication
2. Automated Priority Score Assignment (P1 to P4)
3. Hash Table Indexing + Priority Heap Queue Logic
4. CSV File Persistence so records reload after program restart
5. Role Menus for Student filing, Warden allocation, and Staff resolution

Key Features

> Priority scoring based on issue category (Electrical, Plumbing, Cleaning)

> O(1) Hash Table lookup by Student Roll No and Complaint ID

> Priority Queue sorts urgent hazards to the front for repair staff

> FIFO Queue handles standard room cleaning requests in submission order

> Stack undo log to track and revert recent warden status changes

Algorithms Used

Algorithm 1 - Priority Handling

1. Check issue category & severity level.
2. Assign numerical priority score.
3. Insert node into Priority Heap Queue.
4. Pop highest urgency complaint first.

Algorithm 2 - Hash Table Lookup

1. Take Student ID or Complaint ID.
2. Calculate hash index with key % size.
3. Retrieve record node in O(1) time.
4. Update complaint status in CSV file.

Algorithm 3 - Dijkstra Shortest Path

1. Represent hostel floors as weighted graph.
2. Set room location as start vertex.
3. Calculate shortest path to staff desk.
4. Dispatch nearest available technician.

By organizing tickets into priority levels (Critical, High, Normal, Low) using heaps and hash tables, the system eliminates delays and ensures hostel emergencies get resolved immediately.

Integration of PBL Course Concepts

Course	Concepts Applied	Project Module
Data Structures in C	Hash Table, Priority Queue (Heap), Queue, Linked List, Stack	Emergency Triage Engine, Routine Request Buffer, Fast ID Indexing, Activity History & Undo Stack
OOPs with C++	Classes, Inheritance, Polymorphism, STL, File I/O, extern "C" Integration	Role Entities (Student/Warden/Staff), Backend Architecture, CSV Persistence

Requirement: Demonstrate meaningful integration of both subjects in one project.

System Architecture / Workflow

Complaint Handling Workflow



CSV stores records and restores data after server restart.

Major Components

- Student Module: Login, complaint submission, and status tracking
- C++ Core Engine: User validation, priority rating, and role access
- DSA Dispatch Engine: Heap priority queue and O(1) hash indexing
- Warden & Staff Console: Task assignment, status updates, and notes
- CSV File Storage: Saves users, complaints, and history across restarts

Data / Control Flow

- Student logs in and submits a room grievance with category details.
- System calculates an urgency score and creates a complaint token.
- Ticket is saved to the CSV file and inserted into the hash table.
- Heap queue pops critical tickets first for warden review.
- FIFO queue handles routine room cleaning in order of filing.
- Staff member completes the repair and enters resolution notes.
- Final status and timestamp are updated in the CSV database.

Technology Stack & Methodology

Technology Stack

- **Programming:** C++17 & Standard C
- **Storage:** Plain CSV text files using std::fstream
- **Tools / Framework:** Standard Template Library, extern "C"
- **IDE & Editors:** VS Code / Code::Blocks / GCC Compiler
- **Version Control:** Git & GitHub Repository

Development Methodology

1. Requirement Analysis (Hostel workflow review)
2. System Design (Class hierarchy & data structures)
3. Implementation (C++ coding & file persistence)
4. Testing (Edge cases, invalid inputs & recovery)
5. Evaluation (Lookup speed & usability demo)

We chose C++ and CSV files to build and control custom data structures directly from scratch without requiring extra database setup or web server overhead.

Initial Progress & Team Contribution

Progress Achieved

- Studied existing hostel ticketing systems and identified drawbacks.
- Finalized user roles: Student, Floor Warden, and Repair Staff.
- Created base C++ classes and core binary heap queue functions.

Individual Contribution

Member	Role	Contribution
Harshit	Lead	Architecture design, Priority Queue heap module
Diviya	Developer	Hash Table indexing, complaint matching portal
Aditya	Testing/Docs	CSV file persistence, edge-case testing & slides

Roadmap, Expected Outcomes & References

Roadmap

1. Phase-I: Requirement analysis & core data structure design
2. Phase-II: Module development & CSV file persistence
3. Phase-III: System testing, edge-case handling & final report

Expected Outcomes

- Working C++ console program for hostel administration
- O(1) ticket retrieval and instant heap prioritization
- Practical and easy-to-use tool for campus wardens

References

[1] ResearchGate, Automated Web-Based Hostel Management & Complaint Tracking System. Available: https://www.researchgate.net/publication/401357924

[2] ResearchGate, Design and Development of Student Hostel Portals. Available: https://www.researchgate.net/publication/389336918

[3] GeeksforGeeks, Binary Heap Priority Queue Applications. Available: https://www.geeksforgeeks.org/dsa/applications-priority-queue/

Thank You

Questions & Suggestions